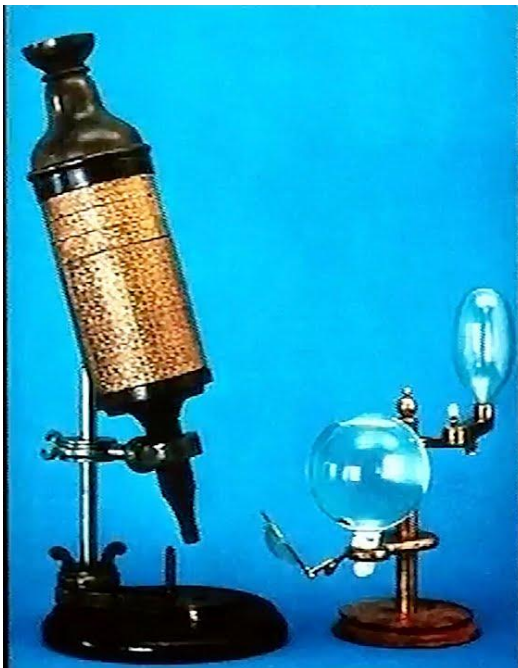


## Michael Y's Summer Biology Course

### Ch 2 Cellular organizations

#### The Cell theory 細胞學說

|                                                         |                                                             |
|---------------------------------------------------------|-------------------------------------------------------------|
| <b>Cells are the smallest functioning units of life</b> | A cell can perform all seven characteristics of life        |
| <b>All organisms consist of one or more cells</b>       | Unicellular Organisms (單細胞) & Multicellular Organisms (多細胞) |
| <b>Cells arise from pre-existing cells</b>              | E.g. Cell division                                          |



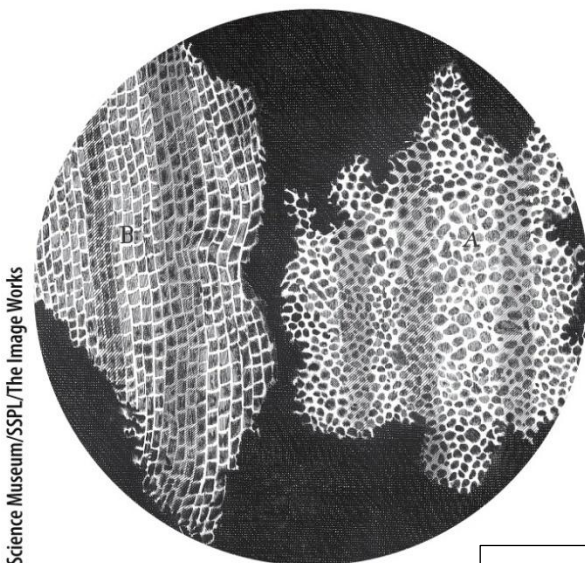
Drawing by Hooke

Robert Hooke 羅拔·虎克

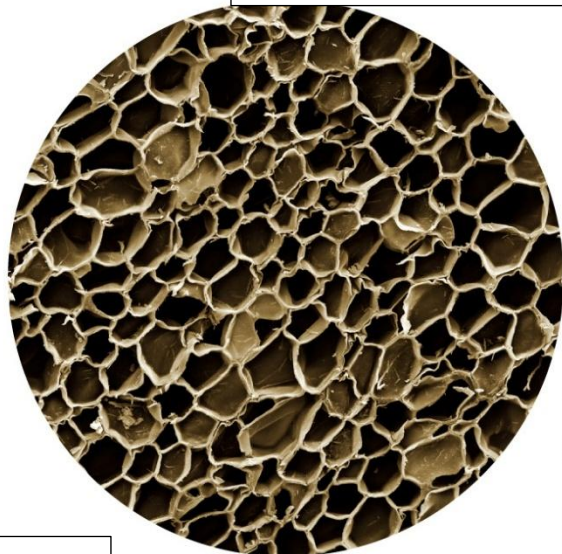


Cork tissue

軟木組織: actually dead cells!



Science Museum/SSPL/The Image Works

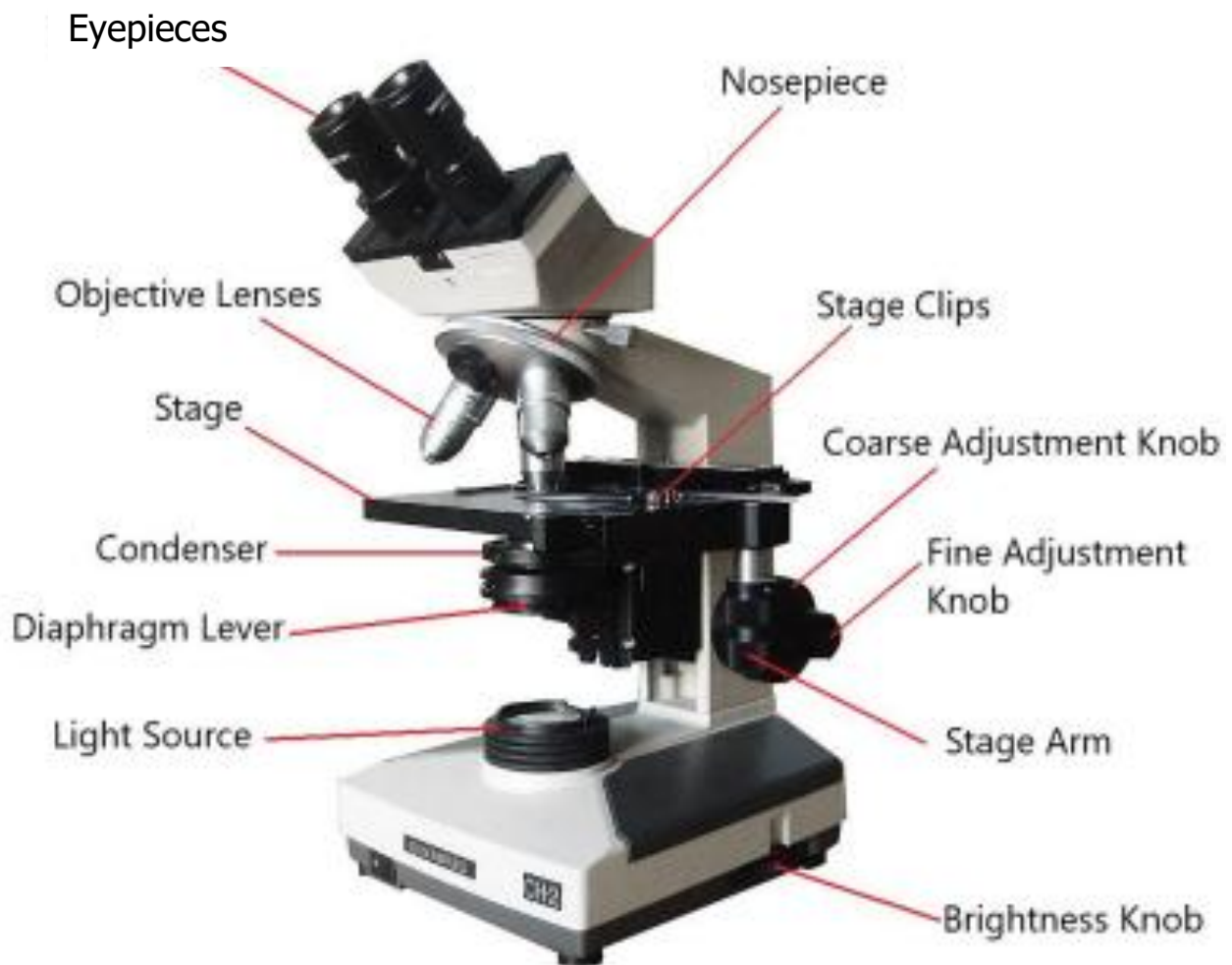


Ted Kinsman/Science Source

Cells “牢房”

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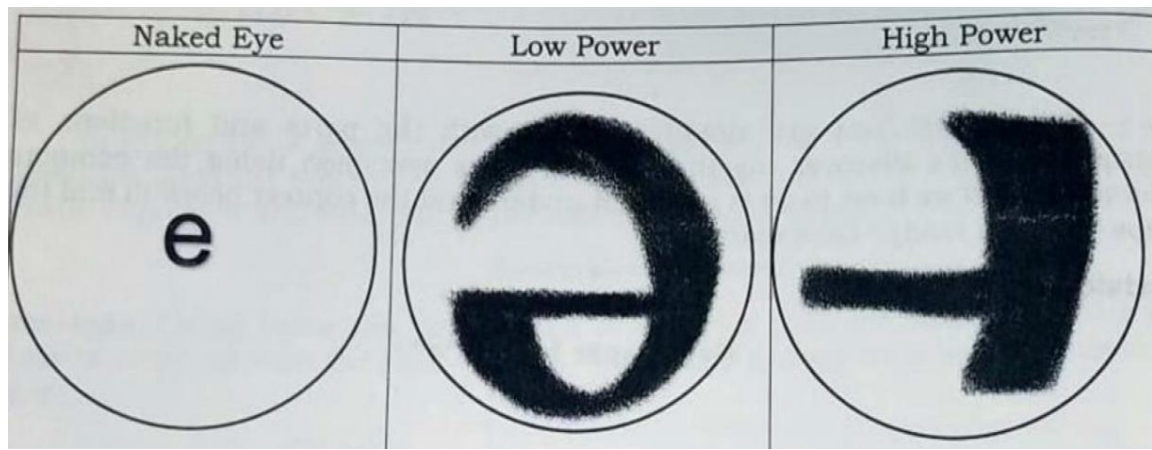
### The modern light microscope 光學顯微鏡



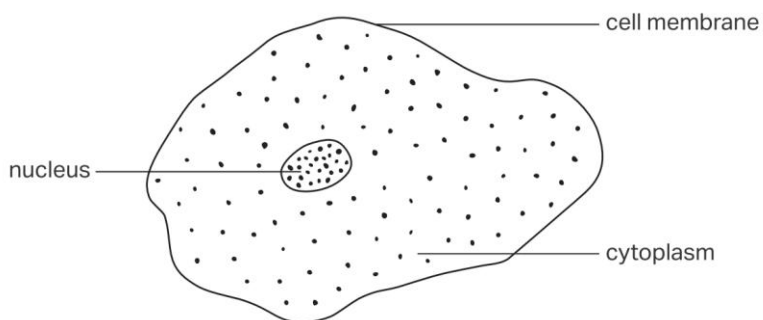
| Parts                                | Function                                             |
|--------------------------------------|------------------------------------------------------|
| <b>Eyepieces (目鏡)</b>                | Magnifies 放大 the image                               |
| <b>Nosepiece (旋轉鏡筒)</b>              | Holds and rotates objective lenses                   |
| <b>Objective Lenses (物鏡)</b>         | Magnifies the image:<br>Longer: higher magnification |
| <b>Stage Clips (載物夾) Stage (載物台)</b> | Holds the slide 載玻片                                  |
| <b>Coarse Adjustment Knob (粗調旋鈕)</b> | Moves stage for basic focus                          |
| <b>Fine Adjustment Knob (微調旋鈕)</b>   | Fine-tunes 微調 for sharp focus                        |
| <b>Condenser (聚光器)</b>               | Focuses light on specimen 標本                         |
| <b>Diaphragm Lever (光圈控制桿)</b>       | Adjusts light amount                                 |
| <b>Light Source (光源)</b>             | Provides light                                       |

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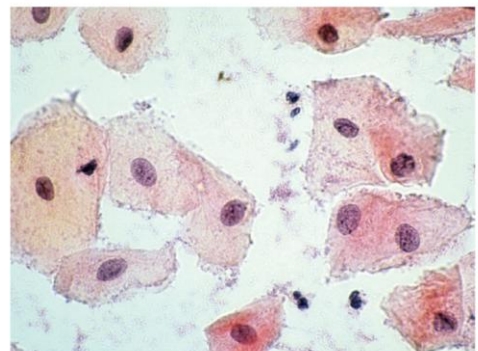
### Inverted image 倒像 under light microscopes



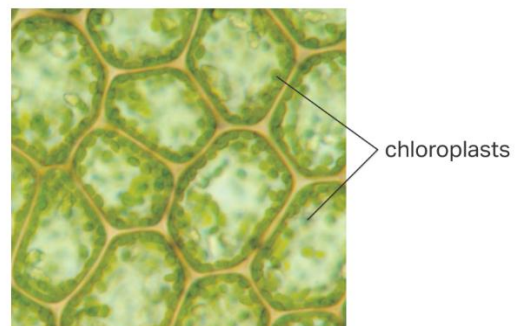
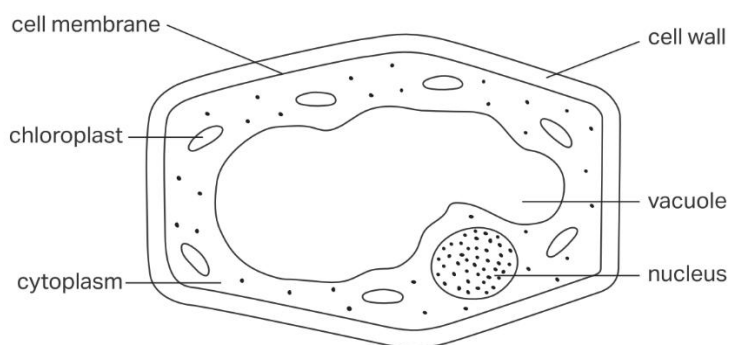
### What you should have learnt about cells in S1 (Ch 4) !



**Fig. 4.4** Drawing of an animal cell



**Fig. 4.5** Mouth epithelial cells in humans (600X)



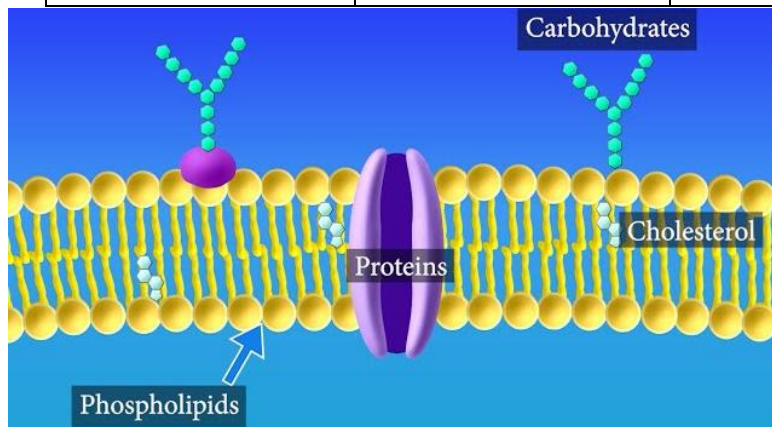
**Fig. 4.7** Plant cells containing chloroplasts (600X)



## Michael Y's Summer Biology Course

### Recalling carbohydrates, lipids and proteins related to cells

|               |                       | Names                                             | Functions                                                                                                                                                                                      |
|---------------|-----------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carbohydrates | Monosaccharides<br>單糖 | Glucose 葡萄糖                                       | Main respiratory 呼吸作用 fuel for quick energy                                                                                                                                                    |
|               | Polysaccharides<br>多糖 | Starch 澱粉                                         | Energy storage in plants                                                                                                                                                                       |
|               |                       | Cellulose 纖維素                                     | Structural component in <b>plant cell wall</b> 細胞壁                                                                                                                                             |
| Lipids        | Phospholipids<br>磷脂質  | Phosphate 磷+<br>Glycerol 甘油+<br>2 fatty acids 脂肪酸 | Main component of <b>cell membranes</b> 細胞膜                                                                                                                                                    |
|               | Steroids<br>類固醇       | e.g. Cholesterol<br>膽固醇                           | Sex hormones 性荷爾蒙<br><b>Help to stabilize 穩定 cell membranes</b>                                                                                                                                |
|               | Protein<br>蛋白質        | Amino acids<br>胺基酸                                | <ul style="list-style-type: none"> <li>Growth and repair 修復 of tissues 組織</li> <li><b>Found in cell membranes</b></li> <li>Enzymes 酶: Speed up reactions</li> <li>Most hormones 荷爾蒙</li> </ul> |

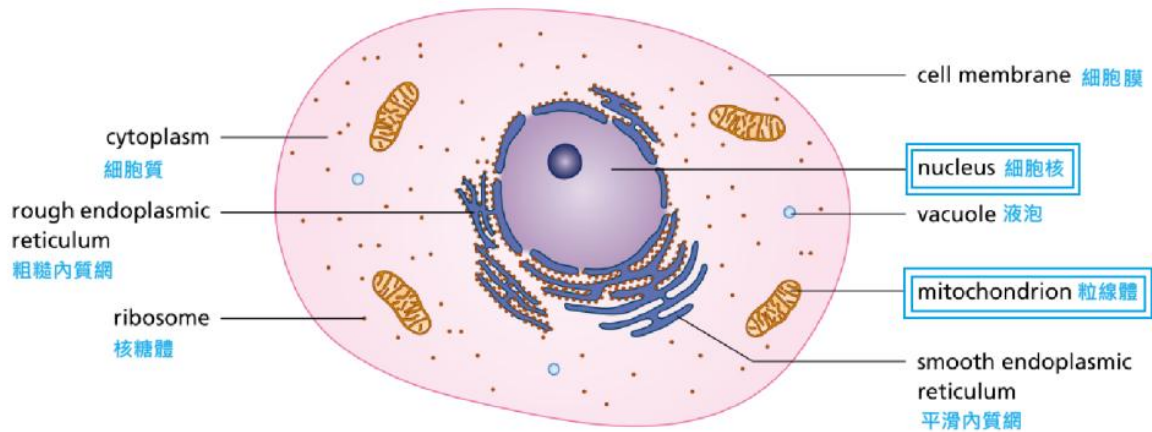


**The cell membrane mainly consists of:**

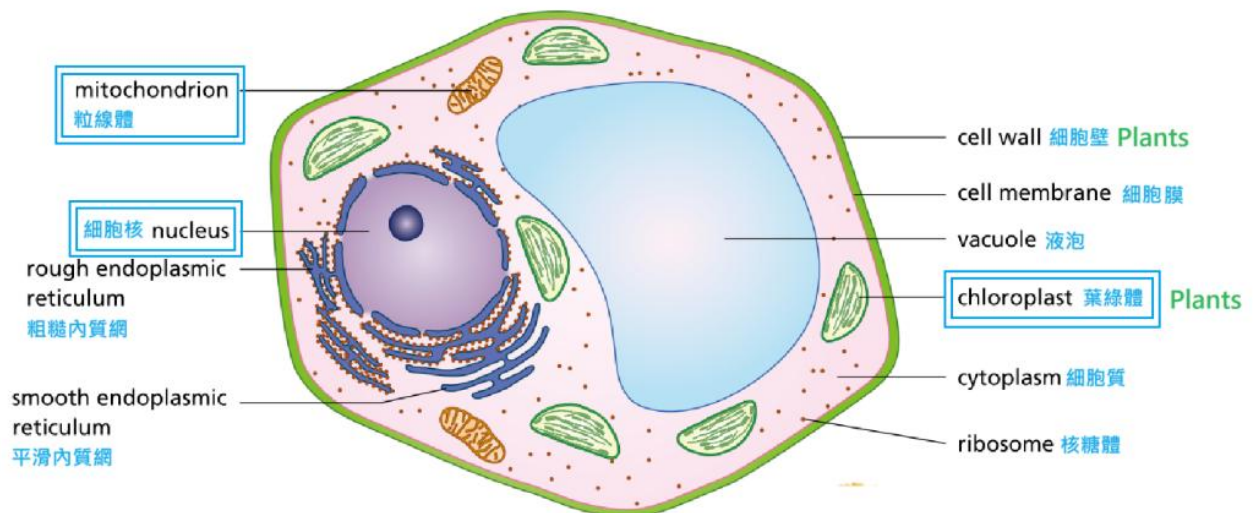
- Phospholipid bilayer** 雙層
- Embedded** 嵌入 **proteins**

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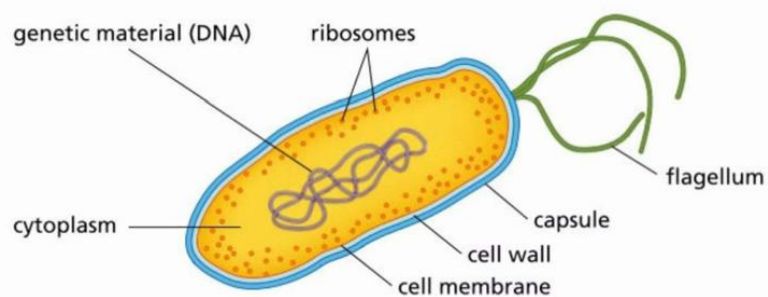
## Animal Cell Structure



## Plant Cell Structure

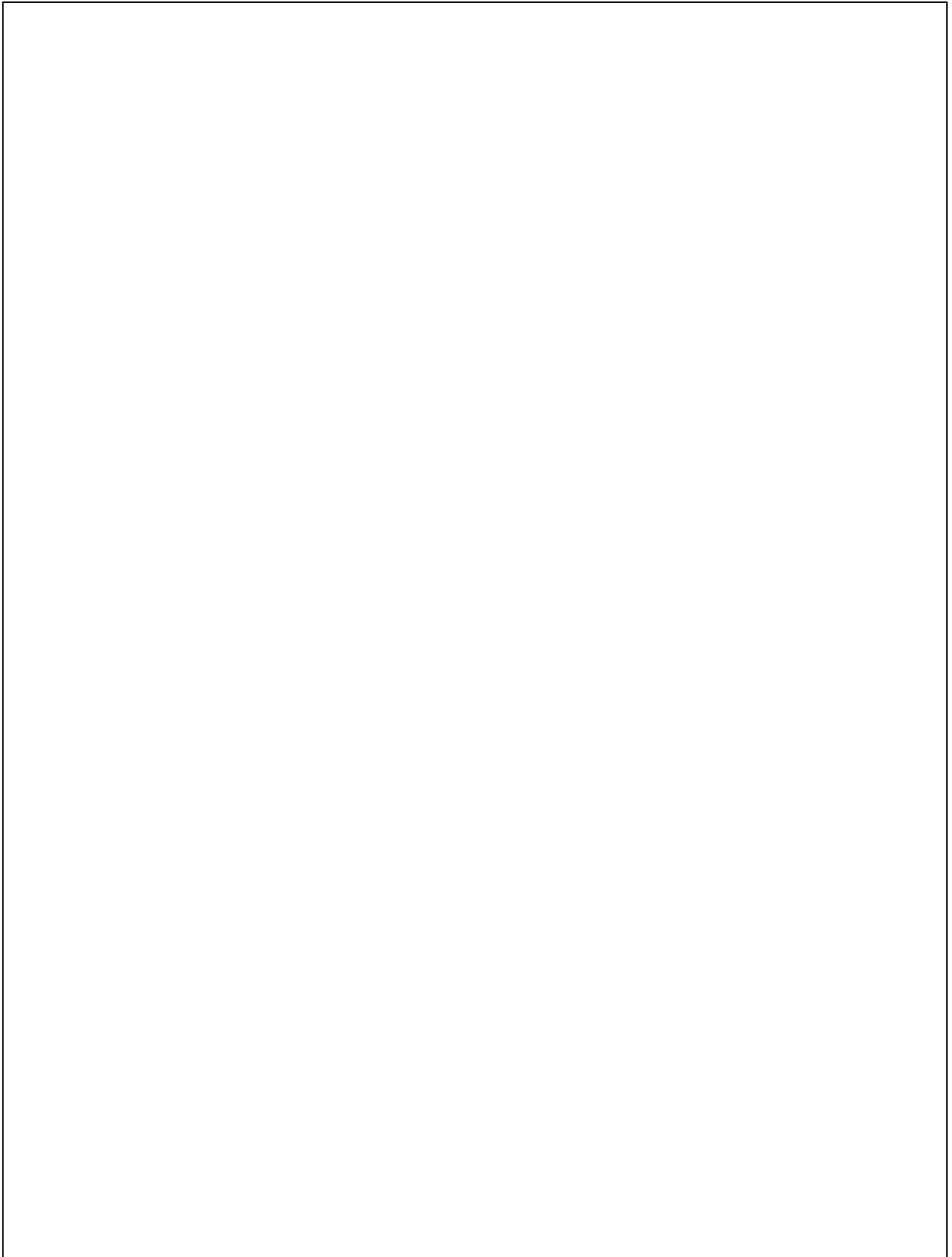


## Prokaryotic cells 原核細胞



## **Michael Y's Summer Biology Course**

**From prokaryotes 原核生物 to eukaryotes 真核生物**



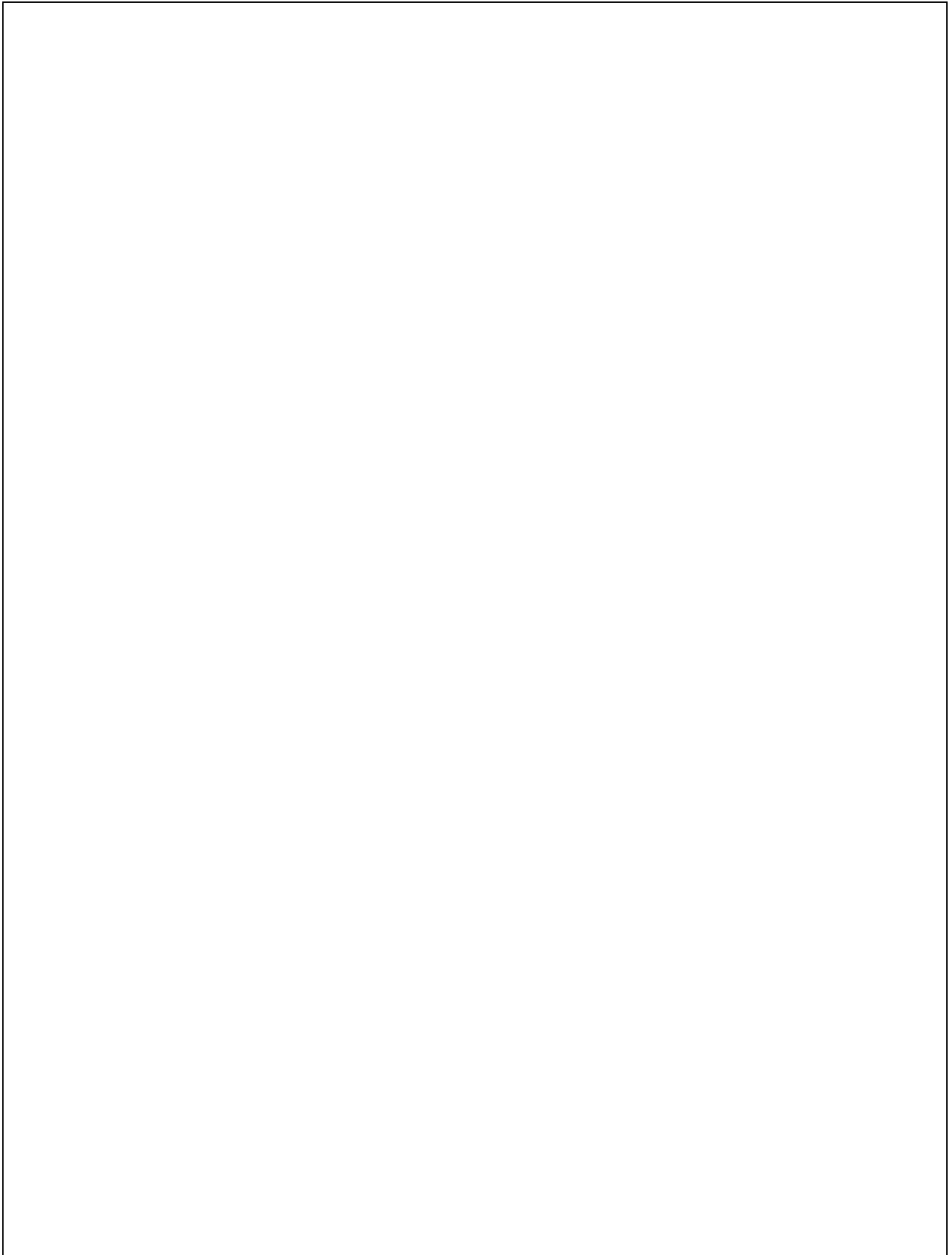
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## Organelles 細胞器

| Organelle                   | Animal | Plant                     | Membrane                          | Details                                                                                                                                                                                                                                                        |
|-----------------------------|--------|---------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cell membrane</b><br>細胞膜 | ✓      | ✓                         | Single                            | <ul style="list-style-type: none"> <li>Made of <b>phospholipid bilayer</b> &amp; <b>proteins</b></li> <li><b>Differentially permeable</b> 差異滲透性</li> <li>Controls substances movement <u>in and out</u> of cells</li> </ul>                                    |
| <b>Cell wall</b><br>細胞壁     | X      | ✓                         | N/A                               | <ul style="list-style-type: none"> <li>Made of <b>cellulose</b> 纖維素 (a carbohydrate)</li> <li>Fully permeable</li> <li>Structural support and protection</li> </ul>                                                                                            |
| <b>Cytoplasm</b><br>細胞質     | ✓      | ✓                         | N/A                               | <ul style="list-style-type: none"> <li>Jelly-like 果凍狀</li> <li>made of water &amp; dissolved substances (e.g. protein)</li> <li>Site for <b>reactions</b></li> </ul>                                                                                           |
| <b>Vacuole</b><br>液泡        | ✓      | ✓<br><u>Large central</u> | Single                            | <ul style="list-style-type: none"> <li>Filled with <b>water + dissolved substance</b> (e.g. minerals)</li> </ul>                                                                                                                                               |
| <b>Nucleus</b><br>細胞核       | ✓      | ✓                         | <u>Double</u><br>nuclear membrane | <ul style="list-style-type: none"> <li>Contain <b>DNA</b> (genetic material)</li> <li><b>Control activities of cell</b><br/><b>(control protein synthesis 合成)</b></li> <li>Nucleolus 核仁: make ribosomes</li> </ul>                                             |
| <b>Ribosomes</b><br>核糖體     | ✓      | ✓                         | N/A                               | <ul style="list-style-type: none"> <li>For <b>protein synthesis</b></li> </ul>                                                                                                                                                                                 |
| <b>Rough ER</b><br>粗面內質網    | ✓      | ✓                         | Single                            | <ul style="list-style-type: none"> <li><b>Protein synthesis and transport</b></li> </ul>                                                                                                                                                                       |
| <b>Smooth ER</b><br>滑面內質網   | ✓      | ✓                         | Single                            | <ul style="list-style-type: none"> <li><b>Lipid synthesis and transport</b></li> </ul>                                                                                                                                                                         |
| <b>Mitochondria</b><br>粒線體  | ✓      | ✓                         | <u>Double</u>                     | <ul style="list-style-type: none"> <li>Aerobic <b>respiration</b> 需氧呼吸 (needs oxygen) for <b>energy</b></li> </ul>                                                                                                                                             |
| <b>Chloroplast</b><br>葉綠體   | X      | ✓                         | <u>Double</u>                     | <ul style="list-style-type: none"> <li><b>Photosynthesis</b> for food (glucose), store <b>starch</b> granules</li> <li>Contain chlorophyll 葉綠素 (green pigment 色素) that absorbs light</li> <li>Green = have chloroplast, non-green = no (e.g. onion)</li> </ul> |

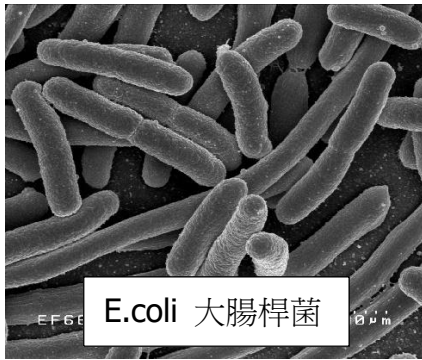
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### **Eukaryotes 真核生物: Animal cells vs plant cells**

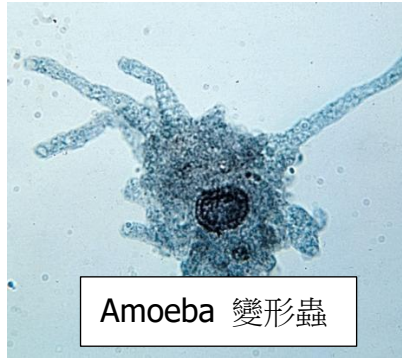




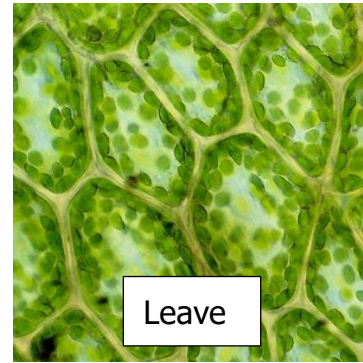
# Michael Y's Summer Biology Course



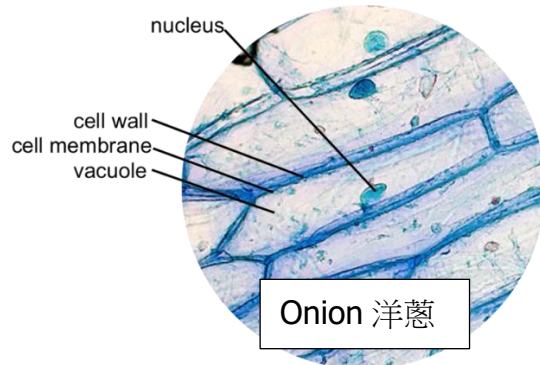
E.coli 大腸桿菌



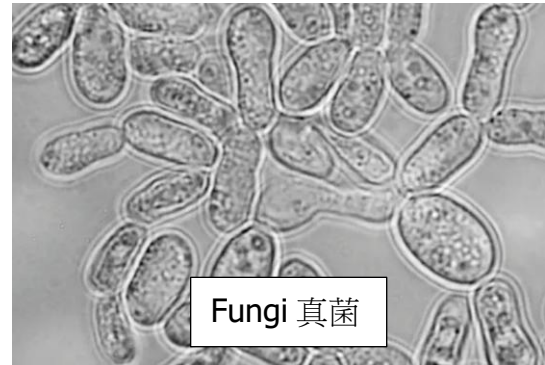
Amoeba 變形蟲



Leave



Onion 洋蔥

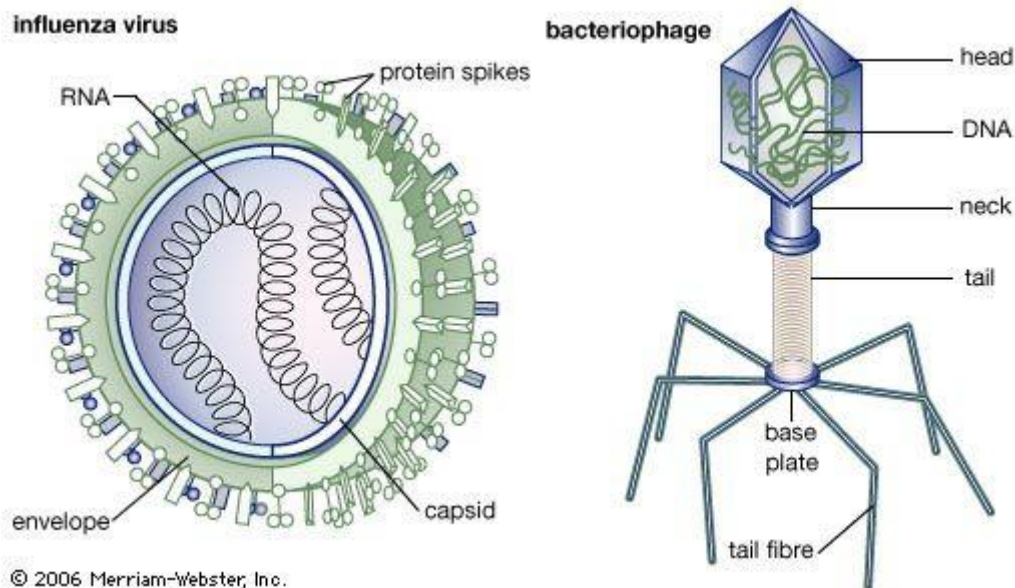


Fungi 真菌

| Feature                             | Prokaryotes<br>原核動物         | Eukaryotes 真核動物                                    |                        |
|-------------------------------------|-----------------------------|----------------------------------------------------|------------------------|
|                                     |                             | Animal Cells                                       | Plant Cells            |
| Examples                            | Bacteria                    | Mushroom, human                                    | Grass                  |
| No. of cells                        | Unicellular<br>單細胞 only     | Unicellular or multicellular 多細胞                   |                        |
| Size                                | Very small                  | smaller                                            | larger                 |
| Shape                               | simpler                     | Irregular                                          | Regular<br>(cell wall) |
| Nucleus<br>細胞核                      | <b>No true nucleus</b>      | <b>With true nucleus 真核</b>                        |                        |
|                                     | DNA lying free in cytoplasm | DNA bounded by nuclear membrane                    |                        |
| Membrane-bound Organelles<br>(膜性胞器) | X                           | ✓<br>(e.g. ER, vacuole, mitochondria, chloroplast) |                        |
| Ribosomes<br>(核糖體)                  | ✓<br>Free in cytoplasm      | ✓<br>Some free, some bound to rough ER             |                        |
| Cell Wall                           | Some                        | <b>Always absent</b><br>(except fungi 真菌)          | <b>Always present</b>  |
| Chloroplasts<br>(葉綠體)               | X                           | X                                                  | ✓<br>(green parts)     |

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### Viruses are not cells! 病毒不是細胞！



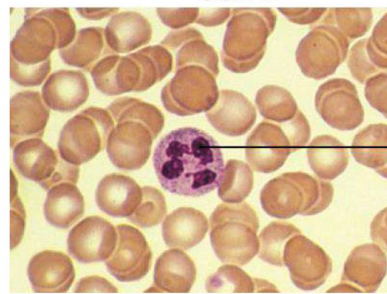
### Concept Check

Cell A



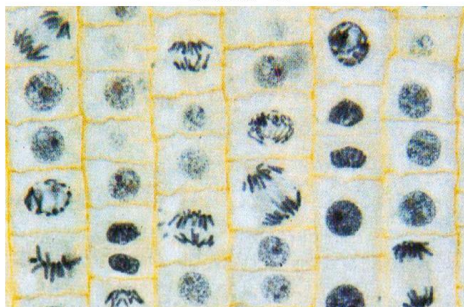
(x150)

Cell B



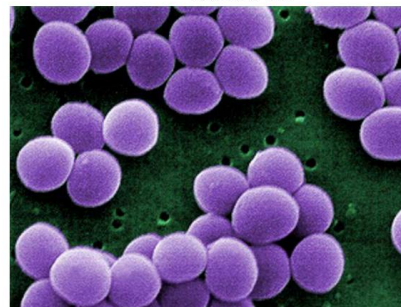
(x1000)

Cells C



(x400)

Cells D



(x20 000)

Identify the above cells:

Prokaryotes:

\_\_\_\_\_

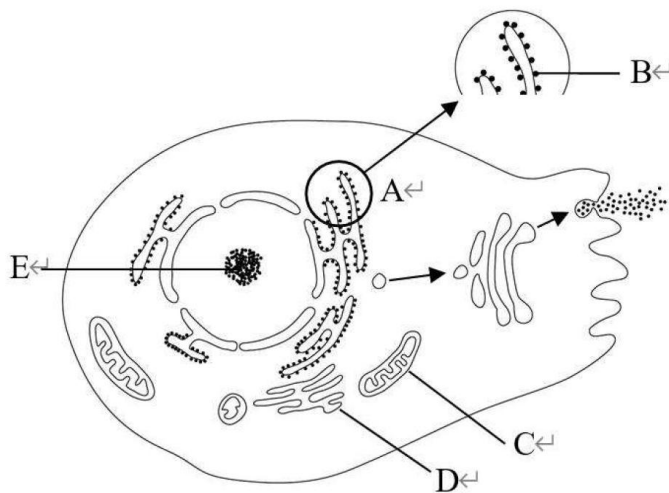
Eukaryotes:

\_\_\_\_\_

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### Multiple Choice Questions (MCQs)

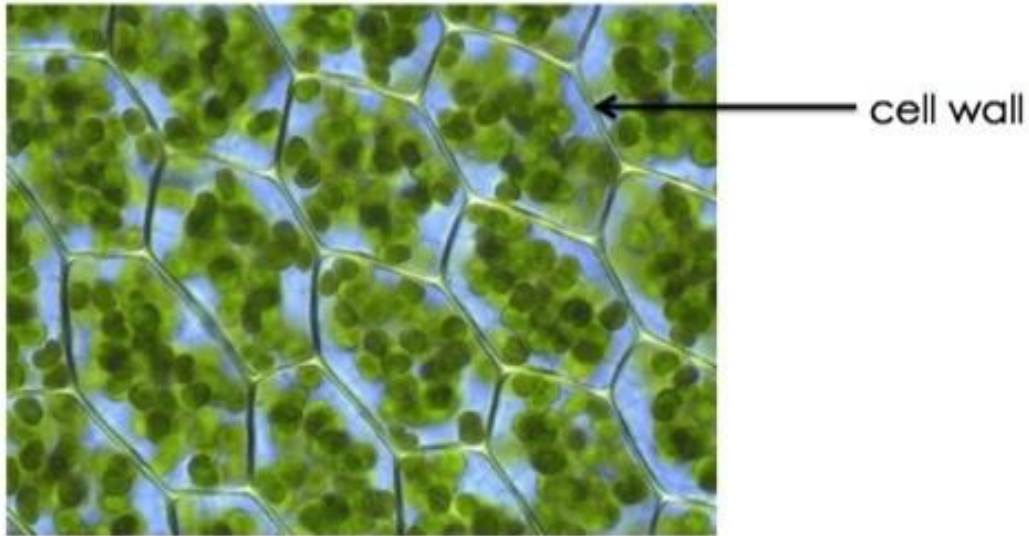
1. Which of the following is the smallest functional unit of life?
  - A. Organ
  - B. Tissue
  - C. Cell
  - D. Organelle
2. Which of the following statements is NOT part of the cell theory?
  - A. Cells are the smallest functioning units of life.
  - B. All organisms consist of one or more cells.
  - C. Cells arise from non-living matter.
  - D. Cells arise from pre-existing cells.
3. Which organelle is responsible for protein synthesis?



- A. Ribosome
- B. Mitochondria
- C. Chloroplast
- D. Vacuole

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4. What is the main component of the cell wall in plants?



- A. Protein
  - B. Cellulose
  - C. Phospholipids
  - D. Starch
5. Which of the following organelles is involved in aerobic respiration?
- A. Nucleus
  - B. Ribosome
  - C. Mitochondria
  - D. Chloroplast
6. Which organelle is absent in animal cells but present in plant cells?
- A. Mitochondria
  - B. Ribosome
  - C. Chloroplast
  - D. Nucleus

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7. What is the function of the vacuole in plant cells?
  - A. Protein synthesis
  - B. Lipid transport
  - C. Storage of water and dissolved substances
  - D. Energy production
  
8. Which feature is present in prokaryotic cells but absent in eukaryotic cells?
  - A. Membrane-bound organelles
  - B. DNA lying free in the cytoplasm
  - C. Ribosomes bound to the ER
  - D. True nucleus
  
9. Which of the following is a characteristic of eukaryotic cells?
  - A. DNA lying free in the cytoplasm
  - B. Absence of membrane-bound organelles
  - C. Presence of a nuclear membrane
  - D. Simpler cell structure
  
10. What is the primary function of the smooth ER?
  - A. Protein synthesis
  - B. Lipid synthesis and transport
  - C. DNA replication
  - D. Photosynthesis

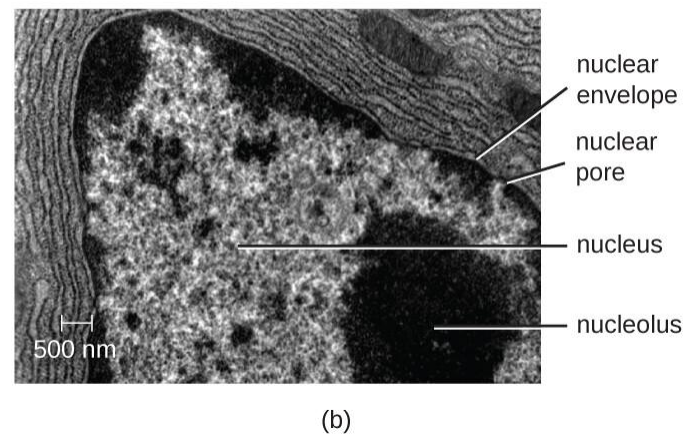
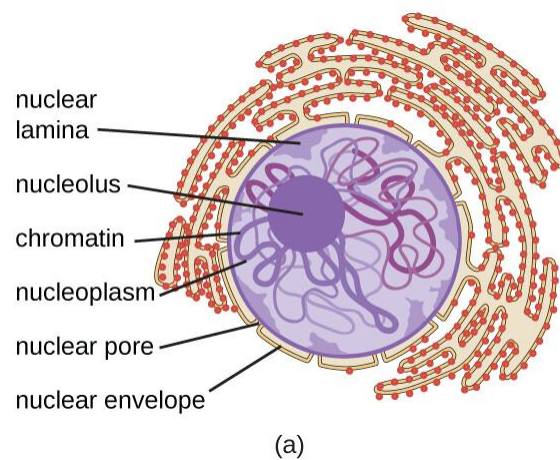


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11. Which of the following is a double-membrane organelle?

- A. Ribosome
- B. Cytoplasm
- C. Mitochondria
- D. Vacuole

12. What is the function of the nucleolus?



- A. Photosynthesis
- B. Protein synthesis
- C. Ribosome production
- D. Lipid transport

13. What is the structural difference between prokaryotic and eukaryotic cells?

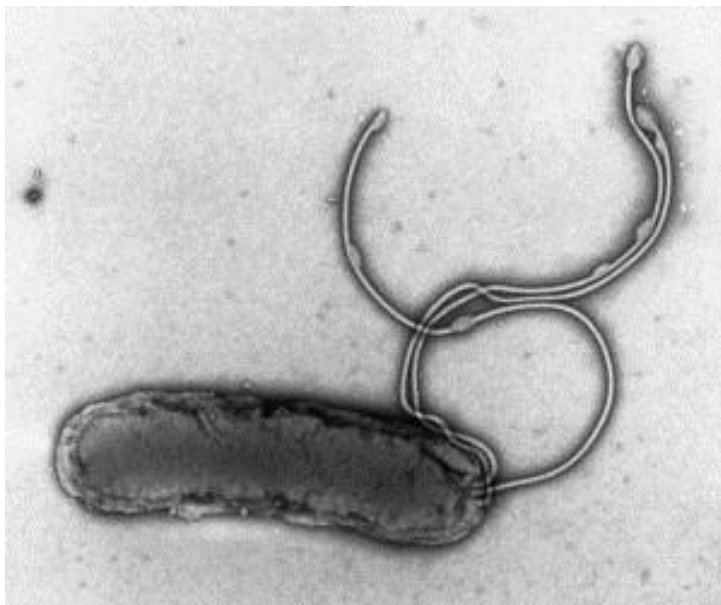
- A. Prokaryotic cells have a true nucleus.
- B. Eukaryotic cells lack membrane-bound organelles.
- C. Prokaryotic cells lack a nuclear membrane.
- D. Eukaryotic cells have no ribosomes.

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14. Which organelle is jelly-like and serves as the site for reactions?

- A. Nucleus
- B. Cytoplasm
- C. Chloroplast
- D. Rough ER

15. Which of the following is NOT found in prokaryotic cells?



- A. Ribosomes
- B. Cell wall
- C. Chloroplasts
- D. DNA

16. What is the main function of the rough endoplasmic reticulum?

- A. Lipid synthesis
- B. Protein synthesis and transport
- C. DNA replication
- D. Energy production

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17. Which of the following is always absent in animal cells?



- A. Mitochondria
- B. Vacuole
- C. Chloroplast
- D. Ribosome

18. What is the main function of the cell membrane?

- A. Energy production
- B. Differentially permeable barrier
- C. Structural support
- D. Photosynthesis

19. Which of the following cells lack a nucleus?

- A. Plant cells
- B. Animal cells
- C. Prokaryotic cells
- D. Fungal cells

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20. What is the main component of the cell membrane?

- A. Cellulose
- B. Phospholipid bilayer and proteins
- C. Starch granules
- D. Chlorophyll

21. What is the primary function of ribosomes?

- A. Lipid transport
- B. Protein synthesis
- C. Energy production
- D. DNA replication

22. Which organelle is responsible for storing genetic material?

- A. Ribosome
- B. Nucleus
- C. Vacuole
- D. Cytoplasm

23. What is the function of the mitochondria?

- A. Photosynthesis
- B. Protein synthesis
- C. Aerobic respiration
- D. Lipid transport

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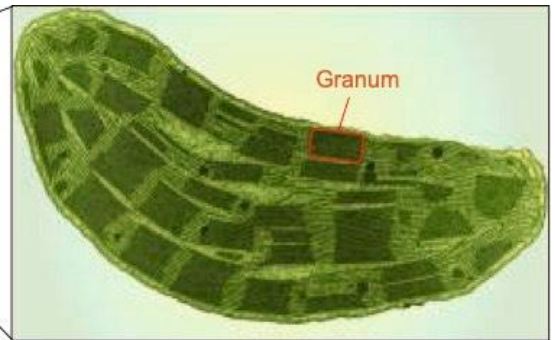
24. Which of the following structures is present in both plant and animal cells?

- A. Cell wall
- B. Chloroplast
- C. Mitochondria
- D. Large central vacuole

25. What is the main function of the chloroplast?



Chloroplasts under **light** microscopy



Chloroplast under **electron** microscopy

- A. Respiration
- B. Photosynthesis
- C. Protein synthesis
- D. Energy storage

26. Which type of cells are simpler in structure?

- A. Prokaryotic
- B. Eukaryotic
- C. Animal cells
- D. Plant cells



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27. What is the shape of plant cells?

- A. Irregular due to cell membrane
- B. Regular due to cell wall
- C. Circular due to cell wall
- D. Oval due to cell membrane

28. Which of the following organelles is single-membraned?

- A. Nucleus
- B. Ribosome
- C. Vacuole
- D. Rough ER

29. Which organelle is made of cellulose?

- A. Cell membrane
- B. Cell wall
- C. Nucleus
- D. Ribosome

30. What is the role of the nuclear membrane?

- A. Protects the nucleus
- B. Synthesizes proteins
- C. Stores water
- D. Produces ribosomes

# **Michael Y's Summer Biology Course**

## **True/False Questions**

**Direction:** put a "✓" if the sentence is correct and put a "X" on the phrase or words that are wrong.

1. Cells are the smallest functioning units of life.
2. All organisms consist of one or more cells.
3. Prokaryotes have membrane-bound organelles.
4. Eukaryotic cells have a true nucleus.
5. The cell wall is made of phospholipid bilayer.
6. Ribosomes are responsible for protein synthesis.
7. Vacuoles are larger and central in plant cells.
8. The mitochondria are involved in aerobic respiration.
9. Chloroplasts are present in animal cells.
10. Cytoplasm is the site for cellular reactions.
11. The nucleus controls the activities of the cell.
12. The nucleolus makes ribosomes.
13. DNA in prokaryotic cells is not enclosed by a nuclear membrane.
14. The rough ER is responsible for lipid synthesis.
15. The cell membrane is fully permeable.
16. The smooth ER is involved in lipid synthesis and transport.
17. Ribosomes are absent in prokaryotic cells.
18. Plant cells have a regular shape due to the cell wall.
19. Mitochondria have a double membrane.
20. Prokaryotic cells are larger than eukaryotic cells.

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### Fill in the Blanks

Choose from the following: (some words could be reused)

- ☐ nucleus 細胞核 ☐ chloroplast 葉綠體 ☐ ribosome 核糖體 ☐ cell wall 細胞壁
- ☐ prokaryotic 原核 ☐ eukaryotic 真核 ☐ cytoplasm 細胞質 ☐ mitochondria 粒線體
- ☐ vacuole 液泡 ☐ cellulose 纖維素 ☐ aerobic respiration 有氧呼吸 ☐ DNA
- ☐ membrane-bound 膜結構

1. \_\_\_\_\_ cells lack a true nucleus.
2. \_\_\_\_\_ cells have a nuclear membrane.
3. The \_\_\_\_\_ controls all cell activities.
4. The \_\_\_\_\_ is the site for protein synthesis.
5. The \_\_\_\_\_ is responsible for aerobic respiration.
6. The \_\_\_\_\_ stores water and dissolved substances in plant cells.
7. The \_\_\_\_\_ is made of cellulose in plant cells.
8. \_\_\_\_\_ cells, like human cells, have membrane-bound organelles.
9. The \_\_\_\_\_ is jelly-like and is the site of reactions.
10. The \_\_\_\_\_ contains genetic material in eukaryotic cells.
11. \_\_\_\_\_ organelles like the rough ER are absent in prokaryotic cells.
12. The \_\_\_\_\_ is responsible for photosynthesis in plant cells.
13. \_\_\_\_\_ is the main component of the plant cell wall.
14. \_\_\_\_\_ is the process that requires oxygen for energy production.
15. \_\_\_\_\_ cells are simpler in structure compared to eukaryotic cells

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## **Answer Key**

### **MCQs:**

1. C
2. C
3. A
4. B
5. C
6. C
7. C
8. B
9. C
10. B
11. C
12. C
13. C
14. B
15. C
16. B
17. C
18. B
19. C
20. B
21. B
22. B
23. C
24. C
25. B
26. A
27. B
28. D
29. B
30. A

### **True/False:**

1. ✓
2. ✓
3. ✗ (Prokaryotes lack membrane-bound organelles.)
4. ✓

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5. ✗ (The cell wall is made of cellulose.)
6. ✓
7. ✓
8. ✓
9. ✗ (Chloroplasts are absent in animal cells.)
10. ✓
11. ✓
12. ✓
13. ✓
14. ✗ (The rough ER is responsible for protein synthesis.)
15. ✗ (The cell membrane is differentially permeable.)
16. ✓
17. ✗ (Ribosomes are present in prokaryotic cells.)
18. ✓
19. ✓
20. ✗ (Prokaryotic cells are smaller than eukaryotic cells.)

### **Fill in the Blanks:**

1. Prokaryotic
2. Eukaryotic
3. Nucleus
4. Ribosome
5. Mitochondria
6. Vacuole
7. Cell wall
8. Eukaryotic
9. Cytoplasm
10. DNA
11. Membrane-bound organelles
12. Chloroplast
13. Cellulose
14. Aerobic respiration
15. Prokaryotic